Report

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2026 / 05 / 15

Summary

Tasks Status

Task	Status	
2026-06 RETINHA presentation	◔	Link
2026-06 Scientific report	◔	Link
2026-11 Journal Article	○	Link
2026-12 Scientific report	○	Link
Dissertation	○	Link

Task	Status	
Journal Club - Automations	◑	Link

Tasks Status

Task	**Status**	**Link**
Model correlated uncertainty in Scaled Spectra		Notebook
↪ Check uncertainty prediction in Atlas experiment ins1759506	◷	
↪ Check effect on pt scaled spectra (Thiago's work)	◷	

Learned Concepts

Initial conditions influence predicted Shear viscosity in heavy ion collision models [1]	✓
Uncertainties in Initial conditions are part of systematic uncertainties [1]	✓

Recent studies

Articles	Status	Link
Bayesian estimation of the specific shear and bulk viscosity of quark–gluon plasma	✓	Bernhard, Nature, 2019

Thesis	Status	Link
Bayesian parameter estimation for relativistic heavy-ion collisions	✓	Bernhard, PhD, 2018

Selected Article

Bayesian inference of the magnetic component of quark-gluon plasma

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(Dated: October 21, 2025)

The chromo-magnetic monopoles (CMM), emergent topological excitations of non-Abelian gauge fields carrying chromo-magnetic charge, have long been postulated to play an important role in the vacuum confinement of quantum chromodynamics (QCD), the deconfinement transition at temperature $T_c \approx 160$MeV, as well as the strongly coupled nature of quark-gluon plasma (QGP). While such CMMs have been found to provide solutions for challenging puzzles from heavy-ion collision measurements, they were typically introduced as model assumptions in the past. Here we show how their very existence can be determined and their abundance extracted in a data-driven way for the first time. Using the CUJET3 framework for calculations of jet energy loss and analyzing a comprehensive experimental data set for nuclear modification factor (R_{AA}) and elliptic flow (v_2) of high-transverse-momentum hadrons, the fraction of CMMs in the QGP is obtained by Bayesian inference and is found to be substantial in the $1 \sim 2T_c$ region. The posterior CMM fraction is further validated by excellent agreement with additional data and is also shown to predict QGP transport properties quantitatively consistent with the state-of-the-art knowledge.

Figure: Statistically rigorous, data-driven extraction of the CMM fraction in QGP, obtained by Bayesian inference, shows that it is substantial in the 1 to 2Tc region.

Results

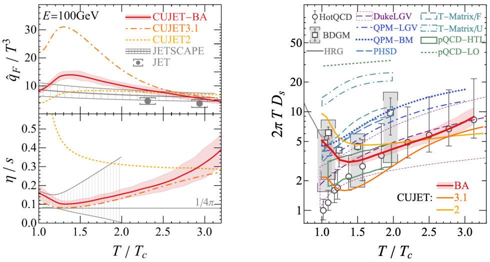

FIG. 4. Transverse-momentum-transfer-squared per unit path length (upper left), shear viscosity (lower left), and diffusion parameter (right) are shown as functions of temperature. In all sub-figures, red lines and bands respectively represent the median and 95% credibility interval of the current Bayesian analysis, and orange dash-dotted (gold dotted) lines corresponds to previous CUJET3.1 (CUJET2) simulations with (without) chromo-magnetic degrees of freedom. Results from other models are included for comparison: JET collaboration model-combined extraction of $\hat{q}$ [84]; JETSCAPE Bayesian analyses of $\hat{q}$ based on energetic particles [68, 69] and η/s for soft particles [71]; diffusion parameter computed from lattice QCD simulations [85–87], UrQMD's simulation based on hadron resonance gas(HRG) model [88], Duke's Langevin-based model [89, 90], Catania's quasi-particle models(QPM) for solving the Boltzmann(BM) and Langevin(LGV) equations [91], PHSD [92–94], TAMU's T-Maxtrix calculation using free-energy(F) and inner-energy(U) assumptions [95–97], and AMY hard thermal loop(HTL) [98] and SUBATECH's leading order(LO) [99, 100] perturbative QCD calculations.

Figure: Results compared between frameworks.

References I

[1] Jonah E. Bernhard. "Bayesian parameter estimation for relativistic heavy-ion collisions". en. arXiv:1804.06469 [nucl-th]. PhD thesis. arXiv, Apr. 2018. URL: http://arxiv.org/abs/1804.06469 (visited on 12/12/2025).